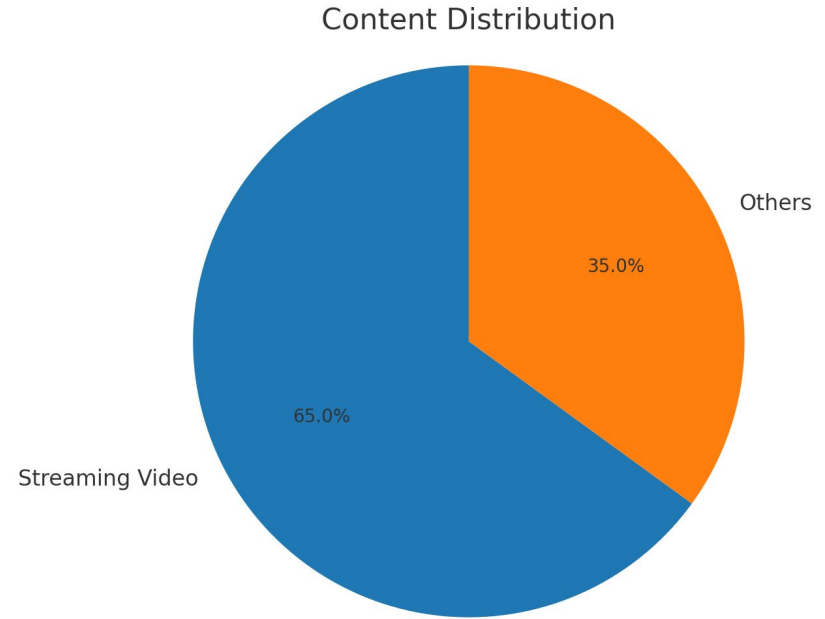


HTTP Live Streaming (HLS) - Brief Overview

Bhargav

Why HLS?

- 65% of global internet traffic is streaming video
- Most popular format of video streaming



History of HLS

- 2009: Apple introduces HLS with iOS 3.0.
- Needed plugin-free streaming over HTTP for mobile networks.
- Designed for CDN scalability and adaptive bitrate



Why Not Use Existing Protocols

- RTMP (Flash)
 - plugin-based
 - security
 - battery concerns.
- RTSP
 - poor CDN scalability
 - firewall issues.
- MSS
 - platform-tied ecosystems.

Turning Point – April 29, 2010

- Steve Jobs writes open-letter 'Thoughts on Flash'
 - Declares that iPhones will never support Flash.
- Guides industry pivot towards HTML5 + HTTP-based streaming

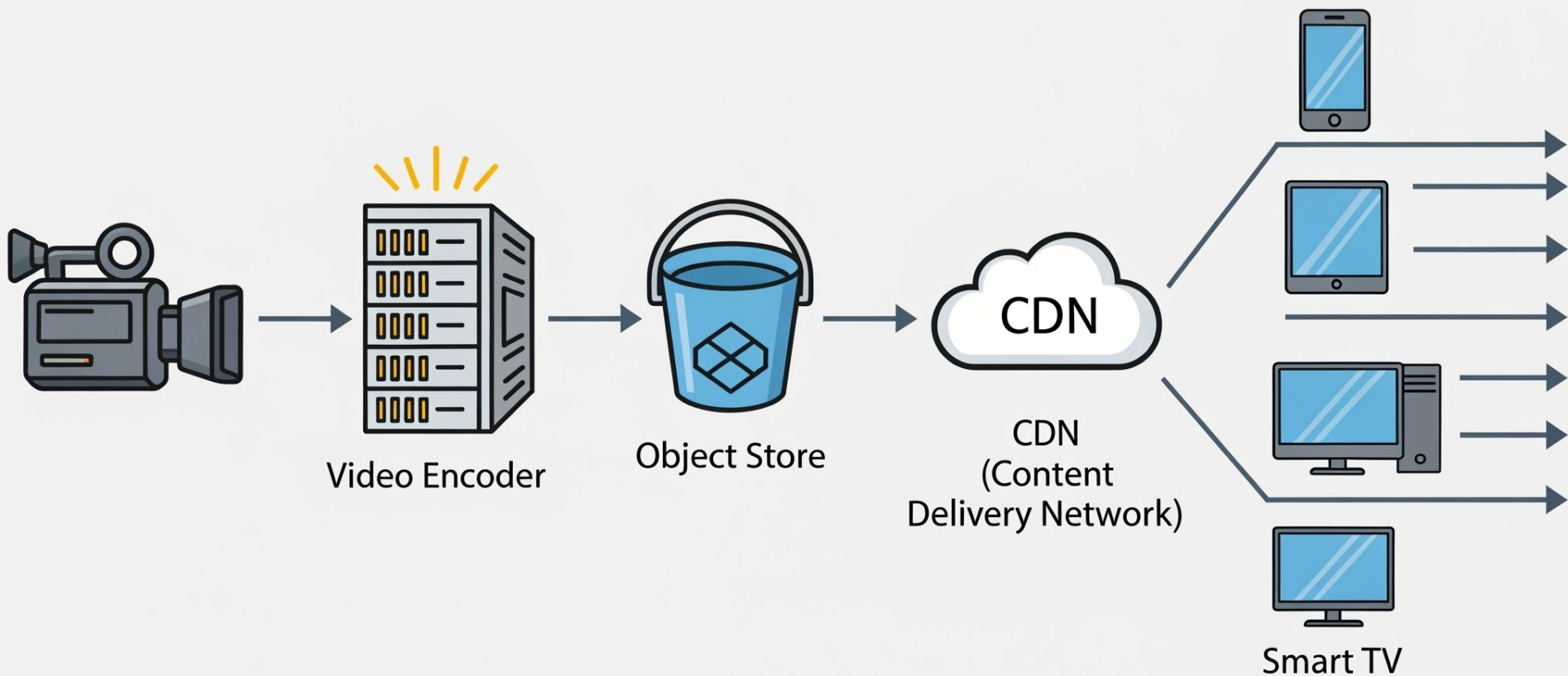
“We know first hand that Flash is the number one reason Macs crash.”

“Flash was designed for PCs using mice, not for touch screens using fingers.”

“By almost any definition, Flash is a closed system.”

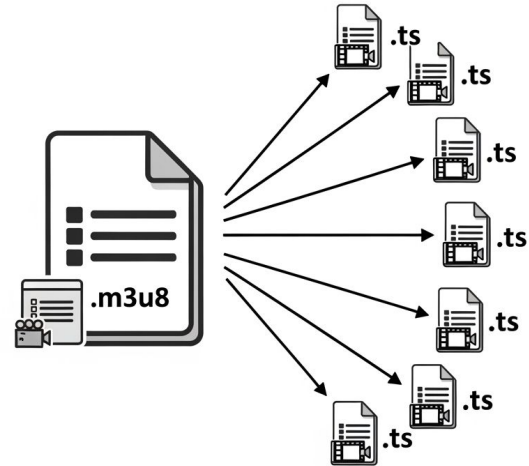
HLS

- Adaptive bitrate streaming protocol
- Video stream is broken down into a series of small, HTTP-based media file segments
 - Streaming problem converted into a file serving problem.
- Firewall Friendly
 - Indistinguishable from regular web traffic
- Widespread Compatibility
 - Standard HTTP - compatible with nearly all modern devices, web browsers, and platforms.



Building Blocks

- Media Segments
 - text-based file, acts as the central index for a video stream
 - does not contain any video data by itself
- Playlist / Manifests
 - small, sequential chunks of the video stream
 - one sequence per playable stream



Media Segments

	MPEG-2 Transport Stream (TS) Segments	Fragmented MP4 (fMP4) Segments
File Extension	.ts	.m4s
Container Structure	Packet-based, self-contained	
Initialization	Self-initializing	Box-based, requires init segment
Overhead	Higher	Requires separate initialization segment
	Higher	Lower
Codec Support	H.264/AAC	Mandatory for HEVC/AV1

HLS Manifest Example

```
#EXTM3U
#EXT-X-VERSION:3
#EXT-X-TARGETDURATION:10
#EXT-X-MEDIA-SEQUENCE:123
#EXT-X-PLAYLIST-TYPE:LIVE
```

```
#EXTINF:10.0,
segment123.ts
#EXTINF:10.0,
segment124.ts
#EXTINF:10.0,
segment125.ts
#EXTINF:10.0,
segment126.ts
```

An abstract graphic featuring two smooth, intersecting curves on a solid blue background. One curve is a light blue color, and the other is a reddish-orange color. They cross each other twice, forming a central lens-like shape. The word "amagi" is centered within this lens in white text.

amagi